

Alvaro Perdomo Gutierrez

Senior Software Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer with **12+** years across logistics, healthcare, and fintech, architecting data pipelines and ML platforms using Python, Azure Databricks, and Apache Airflow. Experienced in ETL/ELT design, data modeling, distributed processing, and governance-first analytics. Proven track record scaling platforms to millions of users while maintaining data quality and performance.

SKILLS

- Languages:** Python, SQL, JavaScript, TypeScript, Java, HTML, CSS
- Data & ML:** Apache Airflow, Azure Databricks, PySpark, Pandas, NumPy, Spark SQL, MLflow, scikit-learn
- Backend:** FastAPI, Django, Node.js, Express, gRPC
- Frontend:** React, Next.js, Vue.js, Tailwind CSS
- Database:** Azure SQL Database, PostgreSQL, MongoDB, TcaplusDB, Elasticsearch, Redis
- Testing:** Pytest, Jest, Selenium, k6
- Cloud & DevOps:** Azure Data Factory, Azure, Google Cloud Platform, Kubernetes, Docker, GitHub Actions, Jenkins, Terraform
- Others:** Power BI, GitHub, RESTful API, GraphQL, Apache Kafka, RabbitMQ, Jira, Confluence, Git

EXPERIENCE

Leanware May 2024–Present
Senior Software Engineer

- Led the design and implementation of data pipeline infrastructure for a logistics and traffic management platform, leveraging **Python**, **Apache Airflow**, and **Azure Data Factory** to ingest, transform, and model real-time GPS and routing data from multiple vehicle sources for analytics consumption.
- Architected ETL workflows using **PySpark** and **Azure Databricks** to process terabytes of geolocation events daily, optimizing query performance through denormalization and implementing incremental refresh patterns that reduced batch job duration by **38%** compared to initial full-load designs.
- Designed and documented data governance standards including schema validation, quality checks, and audit logging across the analytics data warehouse, ensuring compliance with logistics industry data retention policies and enabling trusted analytics outputs to **Power BI** dashboards used by operations teams.
- Collaborated with analytics engineers and business stakeholders to develop dimensional models in **Azure SQL Database** supporting 50+ logistics KPIs including delivery time, route optimization, and fleet utilization metrics used for real-time decision-making.

- Implemented automated data quality frameworks using **Python** and **Pytest**, catching data anomalies within 15 minutes of ingestion and reducing downstream report errors by **92%** across 12+ business intelligence dashboards.
- Migrated legacy batch pipelines from on-premise SQL Server to cloud-native **Azure Databricks** workflows, eliminating infrastructure maintenance overhead and improving pipeline reliability from **94%** to **99.7%** uptime through built-in fault tolerance.
- Mentored two junior data engineers on ETL best practices, **GitHub** workflows, and troubleshooting distributed Spark jobs, establishing code review standards that reduced data model rework iterations by **45%** across the team.
- Optimized transformation logic in **Spark SQL** by identifying and eliminating redundant joins and shuffles, cutting job execution time from 2.5 hours to under 40 minutes and reducing cloud compute costs by **\$8,200 monthly**.
- Designed and maintained **Power BI** semantic models and data refresh schedules to support logistics dashboards monitoring 80+ KPIs, enabling operations teams to identify inefficiencies and achieve 12% improvement in on-time delivery rates.
- Implemented incremental CDC (Change Data Capture) patterns using **Azure Data Factory** to synchronize master data from transactional systems, reducing data staleness from 24 hours to 5 minutes and eliminating nightly reconciliation tasks.
- Authored comprehensive data pipeline documentation and runbooks in **Confluence**, reducing incident resolution time by **40%** and enabling the team to self-service troubleshooting without data platform engineer escalation.
- Built custom monitoring dashboards using **Python** and **Azure Monitor** to track pipeline execution metrics, data freshness, and quality scores, surfacing anomalies proactively before they impact downstream consumers.

ThredUp

Mar 2022–May 2024

Senior Software Engineer

- Spearheaded the migration of legacy batch ML pipelines onto **Azure Databricks**, consolidating 12+ separate **PySpark** jobs into a unified, version-controlled framework that reduced deployment time by **60%** and enabled safer experimentation with ML feature engineering.
- Designed and implemented feature engineering pipelines using **Pandas**, **NumPy**, and **Spark SQL** to generate training datasets for resale-pricing and garment-measurement ML models, supporting inference on 100K+ product images daily with sub-100ms latency at scale.
- Collaborated with data science and platform teams to operationalize model lifecycle management using **MLflow**, establishing reproducible training runs and automated model registry promotion that accelerated time-to-production for new models from 2 months to under 3 weeks.
- Built data infrastructure using **Azure SQL Database** and **Elasticsearch** to serve real-time visual search and style recommendations, handling 50K+ concurrent user sessions and maintaining **P99 query latency below 200ms** during peak traffic periods.

- Implemented robust data quality checks and anomaly detection using **Python** and custom **Pytest** suites, catching training data distribution shifts within 4 hours of ingestion and preventing model degradation that would have reduced recommendation relevance by **8-12%**.
- Optimized **PySpark** transformations through partition strategy refinement and broadcast joins, reducing data processing bottlenecks in the recommendation pipeline by **34%** and enabling near real-time personalization for 172M+ catalog items.
- Architected event streaming infrastructure using **Apache Kafka** to ingest user interaction telemetry, product metadata, and transactional events into the analytics warehouse, enabling behavioral ML features and reducing analytics data latency from 12 hours to 90 seconds.
- Designed dimensional data models in **Azure SQL Database** supporting analytics on 275K weekly image tagging operations and 55K+ unique brands, enabling business intelligence reporting that identified a **15% improvement opportunity** in sell-through rate.
- Mentored two junior engineers on **Airflow** DAG development and **GitHub** code review practices, establishing pipelines-as-code standards that reduced human error incidents in production data jobs by **56%** over six months.

Spring Health

Nov 2020–Mar 2022

Software Engineer

- Contributed to scaling the precision-matching ML platform from handling 200K to 2M+ members by optimizing data pipeline efficiency using **Python** and **PySpark**, implementing incremental processing patterns that reduced compute time by **42%** while maintaining clinical matching validity.
- Designed ETL workflows using **Apache Airflow** to ingest and transform member eligibility, clinical assessments, and provider network data from 50+ enterprise customers including Fortune 500 companies like PepsiCo and Instacart, establishing HIPAA-compliant data lineage and governance.
- Built operational ML pipelines using **Python** and **Pandas** to predict member dropout risk and forecast demand for mental health services, generating actionable insights that enabled proactive outreach and improved member retention by **7.3%** across the 2M+ member base.
- Optimized data model architecture in **PostgreSQL** to support the global Care Navigation platform launch across 6 countries with jurisdiction-aware matching logic, handling 18+ localization and compliance rules while maintaining sub-200ms response times for member queries.
- Implemented data quality frameworks using **Pytest** and custom validation rules to monitor inbound eligibility feeds from enterprise data partners, reducing downstream matching errors by **68%** and improving trust scores for clinical outcomes reporting.
- Collaborated with compliance and security teams to architect HIPAA-compliant data storage and ETL patterns using **Azure SQL Database** and encrypted data vaults, enabling the platform to onboard regulated enterprise customers with confidence.
- Maintained and optimized batch jobs processing 2M+ member records nightly using **Spark SQL**, reducing job duration from 8 hours to 3.5 hours and eliminating processing windows that had previously blocked morning analytics report generation.

Tencent Games
Software Engineer

Nov 2014–Nov 2020

- Built scalable backend services using **Java** and **gRPC** for Honor of Kings, a mobile MOBA that scaled from launch to 100M+ daily active users by November 2020, handling 40M+ concurrent player sessions using **TcaplusDB** as the primary distributed data store.
- Implemented telemetry and analytics pipelines using **Python** and **Apache Kafka** to ingest player behavior, in-game events, and performance metrics from millions of concurrent players, enabling data-driven balance changes and feature optimizations that improved player retention by **8.5%**.
- Developed backend systems for PUBG Mobile using **Java** and **Node.js** to support 100-player battle royale matches, implementing anti-cheat detection using kernel-level ACE technology and custom fraud detection algorithms running on server-side log analysis.
- Optimized game server infrastructure on **Google Cloud Platform** using **Kubernetes** and **Docker**, reducing server initialization time by **45%** and improving player session reliability to **99.6%** uptime across production deployments.
- Designed and maintained data schemas in **TcaplusDB** for game state persistence, player progression, and economy systems serving 40M+ daily transactions, ensuring ACID guarantees and sub-100ms query latencies under peak load.
- Contributed to the Tencent Gaming Buddy emulator project by implementing backend compatibility layers in **Java**, enabling PC emulation of mobile games and expanding the addressable player base by **2.1M+ users** within the first 12 months of launch.

EDUCATION

The Hong Kong Polytechnic University
Bachelor of Science in Computer Science

Sep 2010–Nov 2014